Axcel®

ERYTHROMYCIN ES TABLET

MAL20020154AZ

COMPOSITION:
Each tablet contains
Erythromycin Ethylsuccinate
equivalent to Erythromycin 400mg

PRESENTATION:
Orange colour, oval shaped, biconvex film coated tablets embossed with the word "KOTRa" on one side and plain on the other side.

INDICATIONS:
Indicated in the treatment of infections caused by susceptible strains of designated organisms in the diseases listed below: upper and lower respiratory tract infections that caused by *Streptococcus pyogenes* and *Streptococcus pneumoniae*. Pertussis (whooping cough) caused by *Bordetella pertussis*. Skin and soft tissue infections caused by *Streptococcus pyogenes* and *Streptococcus aureus*. Otitis media, erythrasma, listeriosis, diphtheria, primary syphilis in patients allergy to penicillin, prophylaxis of endocarditis.

PHARMACOLOGY:
Mechanism of Action:
Erythromycin and other macrolides bind reversibly to the 50S subunit of the ribosome, resulting in blockage of the transpeptidation or translocation reactions, inhibition of protein synthesis, and hence inhibition of cell growth. Its action is predominantly bacteriostatic, but high concentrations are slowly bactericidal against the more sensitive strains. Because of the ready penetration of macrolides into white blood cells and macrophages there has been some interest in their potential synergy with host defence mechanism in vivo. Its actions are increased at moderately alkaline pH (up to about 8.5), particularly in Gram-negative species, probably because of the improved cellular penetration of the non ionised form of the drug.
Erythromycin has a broad spectrum activity. The pathogenic organisms which are usually sensitive to Erythromycin include:
a) Gram-positive cocci: *Streptococcus pneumoniae, Streptococcus pyogenes, Staphylococcus aureus.*
b) Gram-positive organisms: *Bacillus anthracis, Corynebacterium diphtheriae etc.*
c) Gram-negative cocci: *Neisseria meningitidis, N. gonorrhoeae, Moraxella catarrhalis.*
d) Gram-negative organisms: *Bordetella spp., Pasteurella, Haemophilus ducreyi.* Among the gram-negative anaerobes more than half of all strains of *Bacteroides fragilis* and many Fusobacterium strains are resistant.
Fungi, yeast, and viruses are resistant to erythromycin.

Pharmacokinetics:
Erythromycin is a macrolide antibiotic with a broad and essentially bacteriostatic action against many Gram-positive and to a lesser extent some Gram-negative bacteria, as well as other organisms including mycoplasmas, spirochaetes, chlamydiae and rickettsiae. Peak plasma concentration of erythromycin generally occurs between 1 and 4 hours after administration. Higher peak concentrations may be achieved on repeated administration four times daily. Erythromycin is widely distributed through out body tissues and fluids, although it does not cross the blood-brain barrier well and concentrations in cerebrospinal fluid are low. Relatively high concentrations are found in the liver and spleen. Erythromycin crosses the placenta: foetal plasma concentrations are variously stated to be 5 to 20% of those in the mother. It is distributed into breast milk. Erythromycin is excreted in high concentrations in the bile and 2 to 5% of an oral dose is excreted in the urine. The half-life of erythromycin is usually reported to be roughly in the range 1.5 to 2.5 hours, although this may be somewhat longer in patients with renal impairment.

DOSAGE AND ADMINISTRATION:
Adults: 400mg (1 tablet), 6 hourly or 800mg (2 tablets), 12 hourly. Dosage may be increased up to 4g per day according to the severity of the infection.

CONTRAINDICATION:
Contraindicated in patients with known hypersensitivity to erythromycin, liver disease and hepatic impairment.

PRECAUTION:
Since the erythromycin is principally excreted by the liver, caution should be exercised when administered to patients with impaired hepatic function. There have been reports that erythromycin may aggravate the weakness of patients with myasthenia gravis. Prolonged or repeated use of erythromycin may result in an overgrowth of non susceptible bacteria or fungi. Caution should be also taken when administered to patients with a history of arrhythmias.
Pseudomembranous colitis has been reported under the usage of erythromycin, and may range in severity from mild to life threatening side effects. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of erythromycin.
Rhabdomyolysis with or without renal impairment has been reported in seriously ill patients receiving erythromycin concomitantly with lovastatin. Therefore, patient treated with this combination should be carefully monitored for creatine kinase and serum transaminase levels. There have been reports of infantile hypertrophic pyloric stenosis (IHPS) occurring in infants following erythromycin therapy. In one cohort of 157 newborns who were given erythromycin for pertussis prophylaxis, seven neonates (5%) developed symptoms of non-bilious vomiting or irritability with feeding and were subsequently diagnosed as having IHPS requiring surgical pyloromyotomy. Since erythromycin may be used in the treatment of conditions in infants which are associated with significant mortality or morbidity (such as pertussis or chlamydia) the benefit of erythromycin therapy needs to be weighed against the potential risk of developing IHPS. Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalized exanthematous pustulosis (AGEP)], Erythromycin should be discontinued immediately and appropriate treatment should be urgently initiated.

Use in Pregnancy and Lactation:
Erythromycin should be used during pregnancy only if clearly needed. Erythromycin is excreted in human milk. Caution should be exercised when it is administered to nursing women.

SIDE EFFECTS:
Gastrointestinal disturbances such as: abdominal discomfort and cramp, nausea, vomiting and diarrhoea; pseudomembranous colitis, hepatotoxicity, ototoxicity. Allergic reactions ranging from urticaria to anaphylaxis have occurred. Rarely, erythromycin has been associated with the production of ventricular arrhythmias, including ventricular tachycardia. Other side effects that have been reported in patients receiving erythromycin include agranulocytosis, central neurotoxicity including psychotic reactions and nightmares, myasthenia-like syndrome and pancreatitis.

Skin and Subcutaneous Tissue Disorders
Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Post Marketing Experience:
Gastrointestinal Disorders: Infantile hypertrophic pyloric stenosis.

DRUG INTERACTIONS:
Erythromycin should be given with caution if other hepatotoxic or ototoxic drugs are given concomitantly. Cimetidine might increase the risk of toxicity. The use of erythromycin in patients concurrently taking drugs metabolized by the cytochrome P450 system may be associated with elevation in serum levels of other drugs, such as: carbamazepine, cyclosporine, phenytoin, alfentanil, cisapride, lovastatin, bromocriptine, valproate, terfenadine and astemizole. Toxicity of theophylline, digoxin, ergotamine and midazolam may occur when administered concurrently with erythromycin due to the increase of the serum level and decrease of clearance of the particular drug.

OVERDOSAGE AND TREATMENT:
In case of overdosage, erythromycin should be discontinued. Overdosage should be handled with the prompt elimination of unabsorbed drug and all other appropriate measures should be instituted.
Erythromycin is not removed by peritoneal dialysis or hemodialysis.

STORAGE:
Store below 30°C. Protect from light.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:
Available in blister pack of 10 x 10's and 100 x 10's.

Further information can be obtained from pharmacist, physician or the manufacturer.

KOTRa PHARMA
Kotra Pharma (M) Sdn. Bhd. (90082-V)
No.1, 2 & 3, Jalan TTC 12,
Cheng Industrial Estate,
75250 Melaka, Malaysia.

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